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3 Bayes' theorem

Bayes' theorem relates the probability of a hypothesis given the evidence to the probability of the evidence given the hypothesis:

$$P(H \mid E) = P(E \mid H)\, P(H) / P(E)$$

Example 3.1. A test for a rare condition is 99 percent accurate, and one person in a thousand has the condition. If a person tests positive, the chance that they have it is still under ten percent, because false positives from the healthy majority outnumber the true positives.